② Maximize $z = x - y^2$

Subject to $x^2 + y^2 \leq 4$

and $x, y \geq 0$

The Lagrangian function is

$$\bar{z} = x - y^2 + \lambda(4 - x^2 - y^2)$$

The kuhn-Tucker conditions are

$$\frac{\partial z}{\partial x} = 1 - 2x\lambda \leq 0 \qquad \bar{x} \geq 0 \qquad \text{and } \bar{x}\frac{\partial z}{\partial x} = 0$$

$$\frac{\partial z}{\partial y} = -2y - 2y\lambda \leq 0 \qquad \bar{y} \geq 0 \qquad \text{and } \bar{y}\frac{\partial z}{\partial y} = 0$$

$$\frac{\partial z}{\partial \lambda} = 4 - x^2 - y^2 \geq 0 \qquad \bar{\lambda} \geq 0 \qquad \text{and } \cancel{\frac{\partial z}{\partial \lambda}} = \bar{\lambda}\frac{\partial z}{\partial \lambda} = 0$$

To solve a nonlinear programming problem, the typical approach is one of trial and error.

Step 1: First check the possibility that $\bar{x} > 0$, $\bar{y} > 0$, but $\bar{\lambda} = 0$.

From complementary slackness we can write

$$\frac{\partial z}{\partial x} = 0 \quad [\because \bar{x} > 0] \qquad \text{and} \qquad \frac{\partial z}{\partial y} = 0 \quad [\because \bar{y} > 0]$$

or, $1 - 2x\lambda = 0$

or, $1 = 2x\lambda$

or, $x = \frac{1}{2\lambda}$

$\therefore x = \infty \quad [\because \bar{\lambda} = 0]$

or, $-2y - 2y\lambda = 0$

or, $-2y(1+\lambda) = 0$

or, $-2y = 0$

or, $y = 0$

Here, the value of $x = \infty$ violates the conditions of nonnegativity constraints.

Step 2: Check whether $\bar{x}$ or $\bar{y}$ equal to 0, but $\bar{\lambda} > 0$

if $\bar{x} = 0$, $\bar{y} = 2$ $[\because x^2 + y^2 \leq 4]$

As, $\bar{y} > 0$; $\frac{\partial z}{\partial y}$ must have to be 0.

but here, $\frac{\partial z}{\partial y} = -2y - 2y\lambda = -2(2) - 2(2)\cdot\lambda(\geq 0) \neq 0$

If $\bar{y}=0, \bar{x}=2$, therefore $\frac{\partial z}{\partial x}$ must have to be 0.

but here $\frac{\partial z}{\partial x} = 1-2x\lambda = 1-2\cdot2\cdot\lambda(\lambda>0) \neq 0$

Therefore, these solutions violate the condition.

Step 3: Now check the KTC for $\bar{x}>0, \bar{y}>0$, and $\bar{\lambda}>0$

From complementary slackness

$$\frac{\partial z}{\partial x} = 1-2x\lambda = 0 \quad ---\text{(i)}$$

$$\frac{\partial z}{\partial y} = -2y-2y\lambda = 0 \quad ---\text{(ii)}$$

$$\frac{\partial z}{\partial \lambda} = 4-x^2-y^2 = 0 \quad ---\text{(ii)}$$

From equation (i) and (ii) we get

$1-2x\lambda = 0$ and $-2y-2y\lambda=0$

or, $1 = 2x\lambda$ or, $-2y(1+\lambda)=0$

$\therefore \bar{x} = \frac{1}{2\lambda}$ or, $-2y=0$

or, $\bar{y}=0$

Substituting the value of $\bar{x}$ and $\bar{y}$ into (iii) we get

$$4-\left(\frac{1}{2\lambda}\right)^2-(0)^2=0$$

$$\text{or, } 4-\frac{1}{4\lambda^2}=0$$

$$\text{or, } 4=\frac{1}{4\lambda^2}$$

$$\text{or, } 16\lambda^2=1$$

$$\text{or, } \lambda^2=\frac{1}{16}$$

$$\therefore \bar{\lambda}=\frac{1}{4} \text{ (avoid negative value)}$$

$$\therefore \bar{x}=\frac{1}{2\lambda}=\frac{1}{2\cdot\frac{1}{4}}=\frac{1}{\frac{1}{2}}=2 \text{ and } \bar{z}=2$$

Thus, all the conditions are satisfied and the optimal value are $\bar{x}=2, \bar{y}=0, \lambda=\frac{1}{4}, \bar{z}=2$

(3) Maximize $f(x,y) = xy$

Subject to $x^2+y^2 \le 1$

$x, y \geq 0$

The Lagrangian function is

$$z = xy + \lambda(1-x^2-y^2)$$

The Kuhn-Tucker conditions are

$$\frac{\partial z}{\partial x} = y - 2x\lambda \le 0 \qquad \bar{x} \geq 0 \text{ and } \bar{x}\frac{\partial z}{\partial x} = 0$$

$$\frac{\partial z}{\partial y} = x - 2y\lambda \le 0 \qquad \bar{y} \geq 0 \text{ and } \bar{y}\frac{\partial z}{\partial y} = 0$$

$$\frac{\partial z}{\partial \lambda} = 1-x^2-y^2 \geq 0 \qquad \bar{\lambda} \geq 0 \text{ and } \bar{\lambda}\cdot\frac{\partial z}{\partial \lambda} = 0$$

To solve a non-linear programming problem, the typical approach is one of trial and error.

Step 1: First check the possibility that $\bar{x} > 0$, $\bar{y} > 0$, but $\bar{\lambda} = 0$

From complementary slackness we can write

$$\frac{\partial z}{\partial x} = 0 \; [\because \bar{x} > 0] \quad \text{and} \quad \frac{\partial z}{\partial y} = 0 \; [\because \bar{y} > 0]$$

or, $y - 2x\lambda = 0$

or, $y - 2x(0) = 0$ $[\because \lambda = 0]$

or, $y = 0$

or, $x - 2y\lambda = 0$

or, $x - 2y(0) = 0$

or, $x = 0$

This solution of x and y doesnot violate the condition, but we find the maximum value of the function $= 0$.

Step 2: Check whether $\bar{x}$ or $\bar{y}$ equal to 0, but $\bar{\lambda} > 0$

If $x=0$; $y=1$ $[\because x^2+y^2\leq 1]$, therefore, $\frac{\partial z}{\partial y}$ must have to be 0.

But here, $\frac{\partial z}{\partial y} = x-2y\lambda = 0-2(1)\cdot\lambda(>0) \neq 0$

If $y=0$; $x=1$, therefore, $\frac{\partial z}{\partial x}$ must have to be 0.

but, here $\frac{\partial z}{\partial x} = y-2x\lambda = 0-2(1)\cdot\lambda(>0) \neq 0$

Therefore, these solutions violate the condition.

Step 3: Now check the KTC when $\bar{x}>0$, $\bar{y}>0$, and $\bar{\lambda}>0$

From complementary slackness we can write

$$y-2x\lambda=0 \quad ----(i)$$
$$x-2y\lambda=0 \quad ----(ii)$$
$$1-x^2-y^2=0 \quad ----(iii)$$

From equation (i) and (ii) we get

$y=2x\lambda$ and $x=2y\lambda$

$\therefore \lambda=\frac{y}{2x}$ ---(iv) $\therefore \lambda=\frac{x}{2y}$ ---(v)

From (iv) and (v) we can write

$$\frac{y}{2x}=\frac{x}{2y} \text{ or, } 2x^2=2y^2 \text{ or, } x^2=y^2 \quad ---(vi)$$

Substituting (vi) into (iii) we get

$$1-y^2-y^2=0 \text{ or, } 1=2y^2 \text{ or, } y^2=\frac{1}{2} \text{ or, } \bar{y}=\frac{1}{\sqrt{2}}$$

(avoid negative)

therefore, $\bar{x}=\frac{1}{\sqrt{2}}$ and $\lambda=\frac{y}{2x}=\frac{\frac{1}{\sqrt{2}}}{2\cdot\frac{1}{\sqrt{2}}}=\frac{1}{\sqrt{2}}\cdot\frac{\sqrt{2}}{2}=\frac{1}{2}$

Here, $\bar{x}=\frac{1}{\sqrt{2}}$ and $\bar{y}=\frac{1}{\sqrt{2}}$ satisfy all the conditions and the maximum value of $f(x,y)=\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}=\frac{1}{2}$

Thus, the final solution: $\bar{x}=\frac{1}{\sqrt{2}}$, $\bar{y}=\frac{1}{\sqrt{2}}$, $\bar{\lambda}=\frac{1}{2}$ and $\bar{z}=\frac{1}{2}$

(7) Minimize $C = (x_1-3)^2 + (x_2-4)^2$

Subject to $x_1 + x_2 \geq 4$

and $x_1, x_2 \geq 0$

The Lagrangian function is

$$Z = (x_1-3)^2 + (x_2-4)^2 + \lambda(4 - x_1 - x_2)$$

The Kuhn-Tucker conditions are

$$\frac{\partial Z}{\partial x_1} = 2(x_1-3) - \lambda \geq 0 \qquad \bar{x}_1 \geq 0 \text{ and } \bar{x}_1 \frac{\partial Z}{\partial x_1} = 0$$

$$\frac{\partial Z}{\partial x_2} = 2(x_2-4) - \lambda \geq 0 \qquad \bar{x}_2 \geq 0 \text{ and } \bar{x}_2 \frac{\partial Z}{\partial x_2} = 0$$

$$\frac{\partial Z}{\partial \lambda} = 4 - x_1 - x_2 \leq 0 \qquad \bar{\lambda} \geq 0 \text{ and } \bar{\lambda} \frac{\partial Z}{\partial \lambda} = 0$$

Step1: First check the possibility that $\bar{x} > 0$, $\bar{y} > 0$, but $\bar{\lambda} = 0$.

From complementary slackness we can write

$\frac{\partial Z}{\partial x_1} = 2(x_1-3) - \lambda = 0$ $[\because \bar{x} > 0]$ and $\frac{\partial Z}{\partial x_2} = 2(x_2-4) - \lambda = 0$ $[\because \bar{y} > 0]$

or, $2x_1 - 6 = \lambda$

or, $2x_1 - 6 = 0$ $[\because \bar{\lambda} = 0]$

or, $2x_1 = 6$

$\therefore x_1 = 3$

or, $2x_2 - 8 = \lambda$

or, $2x_2 - 8 = 0$ $[\because \bar{\lambda} = 0]$

or, $2x_2 = 8$

$\therefore x_2 = 4$

Here, constraint was $x_1 + x_2 \geq 4$

So, $3 + 4 = 7 > 4$

Therefore, this solution does not violate the condition.

So, $\bar{x}_1 = 3$, $\bar{x}_2 = 4$, $\bar{\lambda} = 0$, and
$$C = (x_1-3)^2 + (x_2-4)^2$$
$$= (3-3)^2 + (4-4)^2$$
$$= 0$$

Step 2: Check whether $\bar{x}$ or $\bar{y}$ equal to 0, but $\bar{\lambda} > 0$

If $x_1 = 0$, $x_2 = 4$ ($\because x_1 + x_2 \geq 4$), therefore, $\frac{\partial z}{\partial x_2}$ must have to be 0.

but, here, $\frac{\partial z}{\partial x_2} = 2(x_2 - 4) - \lambda = \{2(4-4) - \lambda(>0)\} = \{0 - \lambda(>0)\}$
$\neq 0$

If $x_2 = 0$, $x_1 = 4$, therefore, $\frac{\partial z}{\partial x_1}$ must have to be 0.

but, here $\frac{\partial z}{\partial x_1} = 2(x_1 - 3) - \lambda = \{2(4-3) - \lambda(>0)\} \neq 0$.

Therefore, these solutions violate the condition.

Step 3: Now check the KTC when $\bar{x} > 0$, $\bar{y} > 0$, and $\bar{\lambda} > 0$

From complementary slackness we can write

$$2(x_1 - 3) - \lambda = 0 \quad \cdots (i)$$
$$2(x_2 - 4) - \lambda = 0 \quad \cdots (ii)$$
$$4 - x_1 - x_2 = 0 \quad \cdots (iii)$$

From (i) and (ii) we get

$2(x_1 - 3) = \lambda$ and $2(x_2 - 4) = \lambda$

or, $2x_1 - 6 = \lambda \cdots (iv)$ or, $2x_2 - 8 = \lambda \cdots (v)$

From (iv) and (v) we can write

$$2x_1 - 6 = 2x_2 - 8$$
$$\text{or, } 2x_1 = 2x_2 - 8 + 6$$
$$\text{or, } 2x_1 = 2x_2 - 2 \text{ or, } 2x_1 = 2(x_2 - 1) \text{ or, } x_1 = x_2 - 1 \cdots (vi)$$

Substituting (vi) into (iii) we get

$$4 - (x_2 - 1) - x_2 = 0 \text{ or, } 4 - x_2 + 1 - x_2 = 0$$
$$\text{or, } -2x_2 + 5 = 0$$
$$\text{or, } 2x_2 = 5$$
$$\therefore x_2 = 2.5 \left(\tfrac{5}{2}\right)$$

So, $x_1 = x_2 - 1 = 2.5 - 1$
$= 1.5$

and, $\lambda = 2x_1 - 6 = 2(1.5) - 6 = 3 - 6 = -3$

here, $\bar{\lambda} < 0$ violates the nonnegativity condition. Therefore, the optimal solutions are $\bar{x}_1 = 3$, $\bar{x}_2 = 4$, $\bar{\lambda} = 0$, and $\bar{c} = 0$; which we obtain in step 1.

(5) Minimize $C = 4q_1^2 + 2q_2^2$

Subject to $q_1 + q_2 \geq 6$

and $q_1, q_2 \geq 0$

The Lagrangian function is

$$Z = 4q_1^2 + 2q_2^2 + \lambda\left[6 - q_1 - q_2\right]$$

The Kuhn-Tucker conditions are

$$\frac{\partial Z}{\partial q_1} = 8q_1 - \lambda \geq 0 \qquad \bar{q}_1 \geq 0 \qquad \text{and } \bar{q}_1 \frac{\partial Z}{\partial q_1} = 0$$

$$\frac{\partial Z}{\partial q_2} = 4q_2 - \lambda \geq 0 \qquad \bar{q}_2 \geq 0 \qquad \text{and } \bar{q}_2 \frac{\partial Z}{\partial q_2} = 0$$

$$\frac{\partial Z}{\partial \lambda} = 6 - q_1 - q_2 \leq 0 \qquad \bar{\lambda} \geq 0 \qquad \text{and } \bar{\lambda} \frac{\partial Z}{\partial \lambda} = 0$$

To solve a nonlinear programming problem, the typical approach is one of trial and error.

Step-1: Check the possibility that $\bar{q}_1 > 0$, $\bar{q}_2 > 0$, but $\bar{\lambda} = 0$

From complementary slackness we can write

$$\frac{\partial Z}{\partial q_1} = 0 \quad [\because \bar{q}_1 > 0] \qquad \text{and} \qquad \frac{\partial Z}{\partial q_2} = 0 \quad [\because \bar{q}_2 > 0]$$

or, $8q_1 - \lambda = 0$

or, $8q_1 - 0 = 0$ $[\because \bar{\lambda} = 0]$

or, $8q_1 = 0$

$\therefore q_1 = 0$

or, $4q_2 - \lambda = 0$

or, $4q_2 - 0 = 0$

or, $4q_2 = 0$

$\therefore q_2 = 0$

This solution violates the condition of $q_1 + q_2 \geq 6$.

Step-2: Check whether $\bar{q}_1$ or $\bar{q}_2$ equal to 0, but $\bar{\lambda} > 0$.

If $q_1 = 0$; $q_2 = 6$ $[\because q_1 + q_2 \geq 6]$, then $\frac{\partial z}{\partial q_2}$ must have to be zero

but here $\frac{\partial z}{\partial q_2} = 4q_2 - \lambda = 4(6) - \lambda(70) \neq 0$

If $q_2 = 0$; $q_1 = 6$, then $\frac{\partial z}{\partial q_1}$ must have to be zero.

but here $\frac{\partial z}{\partial q_1} = 8q_1 - \lambda = 8(6) - \lambda(70) \neq 0$

Therefore, these solutions violate the condition.

Step-3: Now check the KTC when $\bar{q}_1 > 0$, $\bar{q}_2 > 0$, and $\bar{\lambda} > 0$

From complementary slackness we can write

$8q_1 - \lambda = 0$ --- (i)

$4q_2 - \lambda = 0$ --- (ii)

$6 - q_1 - q_2 = 0$ --- (iii)

From (i) and (ii) we get

$8q_1 - \lambda = 0$ and $4q_2 - \lambda = 0$

or, $\lambda = 8q_1$ --- (iv) or, $4q_2 = \lambda$ --- (v)

From (iv) and (v) we can write

$8q_1 = 4q_2$ or, $q_2 = 2q_1$ — (vi)

Subsituting (vi) into (iii) we get

$6 - q_1 - 2q_1 = 0$ or, $6 - 3q_1 = 0$ or, $6 = 3q_1$

$\therefore \bar{q}_1 = 2$

So, $\bar{q}_2 = 2q_1 = 2 \times 2 = 4$

This solution satisfies the condition $q_1 + q_2 \geq 6$.

Now, $\lambda = 8q_1 = 8 \times 2 = 16$

and $c = 4q_1^2 + 2q_2^2 = 4(2)^2 + 2(4)^2 = (4 \times 4) + (2 \times 16)$
$= 16 + 32 = 48$

Thus, optimal solutions are: $\bar{q}_1 = 2$, $\bar{q}_2 = 4$, $\bar{\lambda} = 16$, and $\bar{c} = 48$

6

Maximize $U = xy^2$

Subject to $x + y \leq 100$

$2x + y \leq 120$

and $x, y \geq 0$

The Lagrangian function of this maximization problem is

$$Z = xy^2 + \lambda_1[100 - x - y] + \lambda_2[120 - 2x - y]$$

The kuhn-Tucker conditions are

$\frac{\partial Z}{\partial x} = y^2 - \lambda_1 - 2\lambda_2 \leq 0 \qquad \bar{x} \geq 0 \quad$ and $\quad \bar{x}\frac{\partial Z}{\partial x} = 0$

$\frac{\partial Z}{\partial y} = 2xy - \lambda_1 - \lambda_2 \leq 0 \qquad \bar{y} \geq 0 \quad$ and $\quad \bar{y}\frac{\partial Z}{\partial y} = 0$

$\frac{\partial Z}{\partial \lambda_1} = 100 - x - y \geq 0 \qquad \bar{\lambda}_1 \geq 0 \quad$ and $\quad \bar{\lambda}_1\frac{\partial Z}{\partial \lambda_1} = 0$

$\frac{\partial Z}{\partial \lambda_2} = 120 - 2x - y \geq 0 \qquad \bar{\lambda}_2 \geq 0 \quad$ and $\quad \bar{\lambda}_2\frac{\partial Z}{\partial \lambda_2} = 0$

To solve a NLP, the typical approach is one of trial and error.

Step-1: Check the possibility that $\bar{x} > 0, \bar{y} > 0, \bar{\lambda}_1 > 0$, but $\bar{\lambda}_2 = 0$

From complementary slackness we can write

$y^2 - \lambda_1 - 2\lambda_2 = 0$ --- (i)

$2xy - \lambda_1 - \lambda_2 = 0$ --- (ii)

$100 - x - y = 0$ --- (iii)

From (i) and (ii) we get

$y^2 - \lambda_1 = 0 \quad [\because \bar{\lambda}_2 = 0]$ and $2xy - \lambda_1 = 0 \quad [\because \bar{\lambda}_2 = 0]$

or, $\lambda_1 = y^2$ $\qquad$ or, $\lambda_1 = 2xy$

Therefore, we can write

$$y^2 = 2xy$$

$$\therefore y = 2x \quad ---(iv)$$

Substituting (iv) into (iii) we get

$$100 - x - 2x = 0$$

$$\text{or, } 100 - 3x = 0$$

$$\text{or, } 3x = 100$$

$$\therefore x = \frac{100}{3}$$

$$\therefore y = 2\left(\frac{100}{3}\right) = \frac{200}{3}$$

$\therefore x + y = \frac{100}{3} + \frac{200}{3} = \frac{100+200}{3} = \frac{300}{3} = 100$; doesnot violate the condition

and $2x + y = 2\left(\frac{100}{3}\right) + \frac{200}{3} = \frac{200}{3} + \frac{200}{3} = \frac{400}{3} = 133\frac{1}{3} > 120$; violates the condition.

So, there are not the solutions.

Step-2: Check if $\bar{x}$ or $\bar{y}$ equal to 0, but $\bar{\lambda}_1 > 0$ and $\bar{\lambda}_2 > 0$

If $\bar{x} = 0$; $\bar{y} = 100$ [$\because x + y \leq 100$]; therefore, $\frac{\partial z}{\partial y}$ must have to be zero.

but, here $\frac{\partial z}{\partial y} = 2xy - \lambda_1 - \lambda_2 = 2 \times 0 \times 100 - \lambda_1(>0) - \lambda_2(>0) \neq 0$

If $\bar{y} = 0$; $\bar{x} = 100$, therefore, $\frac{\partial z}{\partial x}$ must have to be zero.

but, here, $\frac{\partial z}{\partial x} = y^2 - \lambda_1 - 2\lambda_2 = 0 - \lambda_1(>0) - 2\lambda_2(>0) \neq 0$

Therefore, there are not the solutions.

Step-3: Now check $\bar{x} > 0$, $\bar{y} > 0$, $\bar{\lambda}_2 > 0$, but $\bar{\lambda}_1 = 0$

From complementary slackness we can write

$$y^2 - 2\lambda_2 = 0 \quad \text{—(v)}$$
$$2xy - \lambda_2 = 0 \quad \text{—(vi)}$$
$$2x + y = 120 \quad \text{—(vii)}$$

From, (v) and (vi) we get

$y^2 = 2\lambda_2$ and $2xy = \lambda_2$

or, $\lambda_2 = \frac{y^2}{2}$

$\therefore \frac{y^2}{2} = 2xy$ or, $y^2 = 4xy$ or, $y = 4x$ —(viii)

Substituting (viii) into (vii) we get

$2x + 4x = 120$

or, $6x = 120$

$\therefore \bar{x} = \frac{120}{6} = 20$

$\therefore \bar{y} = 4x = 4 \times 20 = 80$

and, $\bar{\lambda}_2 = 2xy = 2 \times 20 \times 80 = 3200$

and, $U = xy^2 = 20 \times (80)^2$

$= 20 \times 6400$

$= 128,000$

Thus, the optimal solutions are, $\bar{x} = 20$, $\bar{y} = 80$, $\bar{\lambda}_1 = 0$, $\bar{\lambda}_2 = 3200$, and $\bar{U} = 128,000$

7 Minimize $C=(x_1-4)^2+(x_2-4)^2$

Subject to $2x_1+3x_2 \geq 6$

$-3x_1-2x_2 \geq -12$

and $x_1, x_2 \geq 0$

The Lagrangian function for this problem is

$$Z=(x_1-4)^2+(x_2-4)^2+\lambda_1(6-2x_1-3x_2)+\lambda_2(-12+3x_1+2x_2)$$

The kunh-Tucker conditions for this minimization problem are :

$\frac{\partial Z}{\partial x_1}=2(x_1-4)-2\lambda_1+3\lambda_2 \geq 0 \qquad \bar{x}_1 \geq 0 \qquad$ and $\bar{x}_1 \frac{\partial Z}{\partial x_1}=0$

$\frac{\partial Z}{\partial x_2}=2(x_2-4)-3\lambda_1+2\lambda_2 \geq 0 \qquad \bar{x}_2 \geq 0 \qquad$ and $\bar{x}_2 \frac{\partial Z}{\partial x_2}=0$

$\frac{\partial Z}{\partial \lambda_1}=6-2x_1-3x_2 \leq 0 \qquad \bar{\lambda}_1 \geq 0 \qquad$ and $\bar{\lambda}_1 \frac{\partial Z}{\partial \lambda_1}=0$

$\frac{\partial Z}{\partial \lambda_2}=-12+3x_1+2x_2 \leq 0 \qquad \bar{\lambda}_2 \geq 0 \qquad$ and $\bar{\lambda}_2 \frac{\partial Z}{\partial \lambda_2}=0$

To solve a nonlinear programming problem, the typical approach is one of trial and error.

Step-1: first check the possibility that $\bar{x}_1>0, \bar{x}_2>0, \bar{\lambda}_1>0$, but $\bar{\lambda}_2=0$

From complementary slackness we can write

$$2(x_1-4)-2\lambda_1+3\lambda_2=0 \quad ---(i)$$

$$2(x_2-4)-3\lambda_1+2\lambda_2=0 \quad ---(ii)$$

$$6-2x_1-3x_2=0 \quad ---(iii)$$

From (i) and (ii) we get

$2x_1-8-2\lambda_1+3\lambda_2=0$

or, $2x_1-8-2\lambda_1=0 \quad [\because \lambda_2=0]$

or, $2x_1-8=2\lambda_1$

or, $\lambda_1=\frac{2x_1-8}{2} \quad ---(iv)$

and $2x_2-8-3\lambda_1+2\lambda_2=0$

or, $2x_2-8-3\lambda_1=0 \quad [\because \lambda_2=0]$

or, $2x_2-8=3\lambda_1$

or, $\lambda_1=\frac{2x_2-8}{3} \quad ---(v)$

Now, from (iv) and (v) we can write

$$\frac{2x_1-8}{2} = \frac{2x_2-8}{3}$$

or, $6x_1 - 24 = 4x_2 - 16$

or, $6x_1 - 4x_2 = -16 + 24$

or, $6x_1 - 4x_2 = 8$

or, $2(3x_1 - 2x_2) = 8$

$\therefore 3x_1 - 2x_2 = 4$ ---(vi)

From (iii) we can write

$2x_1 + 3x_2 = 6$ ---(vii)

we can write the (vi) and (vii) in matrix for as:

$$\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$\therefore \bar{x}_1 = \frac{\begin{vmatrix} 4 & -2 \\ 6 & 3 \end{vmatrix}}{\begin{vmatrix} 3 & -2 \\ 2 & 3 \end{vmatrix}} = \frac{(4\times3)-(-2)\times6}{(3\times3)-(-2)\times2} = \frac{12+12}{9+4} = \frac{24}{13}$$

and $$\bar{x}_2 = \frac{\begin{vmatrix} 3 & 4 \\ 2 & 6 \end{vmatrix}}{\begin{vmatrix} 3 & -2 \\ 2 & 3 \end{vmatrix}} = \frac{(3\times6)-(4\times2)}{13} = \frac{18-8}{13} = \frac{10}{13}$$

Now, substituting the value of $\bar{x}_1$ into (iv) we get

$$\bar{\lambda}_1 = \frac{2x_1-8}{2} = \frac{2\left(\frac{24}{13}\right)-8}{2} = \frac{\frac{48}{13}-8}{2} = \frac{\frac{48-104}{13}}{2}$$

$$= \frac{\frac{-56}{13}}{2} = -\frac{56}{26} = -\frac{28}{13}$$

The negative value of $\bar{\lambda}_1$ violates the nonnegativity ~~restric~~ condition. So, This is not the solution.

Step-2: Now check the possibility that ~~$\bar{x}_1$ or $\bar{x}_2$ equal to 0,~~ ~~but~~ $\bar{\lambda}_1 > 0$ and $\bar{\lambda}_2 > 0$.

Since $\bar{\lambda}_1 > 0$ and $\bar{\lambda}_2 > 0$, from complementary slackness we can write

$$6 - 2x_1 - 3x_2 = 0 \qquad \text{and} \qquad -12 + 3x_1 + 2x_2 = 0$$

$$\text{or, } 2x_1 + 3x_2 = 6 \qquad \text{or, } 3x_1 + 2x_2 = 12$$

in matrix for we can write

$$\begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$

$$\therefore \bar{x}_1 = \frac{\begin{vmatrix} 6 & 3 \\ 12 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix}} = \frac{12-36}{4-9} = \frac{-24}{-5} = \frac{24}{5} > 0$$

$$\text{and} \quad \bar{x}_2 = \frac{\begin{vmatrix} 2 & 6 \\ 3 & 12 \end{vmatrix}}{\begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix}} = \frac{24-18}{-5} = -\frac{6}{5} < 0$$

Therefore, this solution violate the non-negativity restriction.

Step 3: Now check the possibility that $\bar{x}_1 > 0, \bar{x}_2 > 0, \bar{\lambda}_2 > 0$, but $\bar{\lambda}_1 = 0$

From complementary slackness we can write

$$2(x_1 - 4) - 2\lambda_1 + 3\lambda_2 = 0 \quad \text{--- (viii)}$$

$$2(x_2 - 4) - 3\lambda_1 + 2\lambda_2 = 0 \quad \text{--- (ix)}$$

$$-12 + 3x_1 + 2x_2 = 0 \quad \text{--- (x)}$$

From (viii) and (ix) we ~~ca~~ get

$$2x_1 - 8 + 3\lambda_2 = 0 \quad [\because \lambda_1 = 0]$$

$$\text{or, } 3\lambda_2 = 8 - 2x_1$$

$$\therefore \lambda_2 = \frac{8 - 2x_1}{3} \quad \text{--- (xi)}$$

and

$$2x_2 - 8 + 2\lambda_2 = 0 \quad [\because \lambda_1 = 0]$$

$$\text{or, } 2\lambda_2 = 8 - 2x_2$$

$$\text{or, } \lambda_2 = \frac{8 - 2x_2}{2} \quad \text{--- (xii)}$$

From (xi) and (xii) we get

$$\frac{8-2x_1}{3} = \frac{8-2x_2}{2}$$

or, $16-4x_1 = 24-6x_2$

or $16-24 = 4x_1 - 6x_2$

or, $-8 = 2(2x_1 - 3x_2)$

$\therefore\ 2x_1 - 3x_2 = -4$ --- (xiii)

we can rearrange (x) as

$$3x_1 + 2x_2 = 12$$

in matrix form, $\begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -4 \\ 12 \end{bmatrix}$

$$\therefore\ \bar{x}_1 = \frac{\begin{vmatrix} -4 & -3 \\ 12 & 2 \end{vmatrix}}{\begin{vmatrix} 2 & -3 \\ 3 & 2 \end{vmatrix}} = \frac{(-4)2-(-3)12}{(2\times2)-3(-3)} = \frac{-8+36}{4+9} = \frac{28}{13}$$

$$\therefore\ \bar{x}_2 = \frac{\begin{vmatrix} 2 & -4 \\ 3 & 12 \end{vmatrix}}{\begin{vmatrix} 2 & -3 \\ 3 & 2 \end{vmatrix}} = \frac{24-(-4)3}{13} = \frac{24+12}{13} = \frac{36}{13}$$

and, $\bar{\lambda}_2 = \dfrac{8-2x_1}{3} = \dfrac{8-2\left(\frac{28}{13}\right)}{3} = \dfrac{8-\frac{56}{13}}{3} = \dfrac{\frac{104-56}{13}}{3} = \dfrac{\frac{48}{13}}{3}$

$$= \frac{48}{13}\times\frac{1}{3} = \frac{16}{13} = 1\frac{3}{13}$$

This solution doesnot violate any conditions.

Thus, the optimal solutions are

$$\bar{x}_1 = \frac{28}{13},\ \bar{x}_2 = \frac{36}{13},\ \bar{\lambda}_1 = 0,\ \bar{\lambda}_2 = \frac{16}{13}$$

and $\bar{C} = \left(\frac{28}{13}-4\right)^2 + \left(\frac{36}{13}-4\right)^2 = (2.1538-4)^2 + (2.7692-4)^2$

$= (-1.8462)^2 + (-1.2308)^2 = 3.4085 + 1.5149$

$= 4.9234$